Homework-1

Learning in Hand Cube Rotation

1. Baseline

I trained the baseline policy for the Mjlab-Leap-Left-HandCube-Rotate task using 4096 parallel environments, seed 42 and 5000 PPO learning iterations. With 32 environment steps per iteration, the nominal training budget was 655360000 environment transitions. Training was performed using one NVIDIA A100 GPU on USC CARC. The primary training command was:

uv run python scripts/train.py Mjlab-Leap-Left-HandCube-Rotate --env.scene.num-envs 4096 --agent.max-iterations 5000 --agent.seed 42 --agent.run-name baseline_full_5000

The initial Slurm job reached iteration 3794 before its 8 hour limit. Because checkpoints were saved every 100 iterations, I resumed from model_3700.pt and completed training at iteration 4999/5000. The two W&B runs corresponding to the baseline are the initial baseline run and the completed continuation run.

Under fixed evaluation, the baseline achieved the highest rotation progress and cube-retention rate among the three policies.

Note: To ensure a fair comparison, the baseline and both modified policies were evaluated using identical conditions. Each final model_4999.pt checkpoint was evaluated with 512 parallel environments, evaluation seed 123, a control timestep of 0.05 seconds and 1000 control steps per environment. This corresponds to 512000 environment transitions for each policy.

2. Baseline Evaluation

I evaluated the final baseline checkpoint using 512 parallel environments for 1000 control steps with evaluation seed 123. This provided 512,000 environment transitions and 1091 completed episodes. The baseline achieved a mean rotation progress of 0.2226, a cube retention rate of 93.86% and a mean episode length of 18.82 seconds. Its mean position and tilt errors were 0.01148 and 0.2128 respectively.

Only 6.14% of the baseline episodes ended in failure, with every failure caused by excessive cube pose deviation. There were no cube drop, excessive speed, or NaN terminations. These results indicate that the baseline generally retained the cube and produced stable rotation, although occasional translational or tilt deviation remained its primary failure mode.

Metric	Baseline result
Rotation progress	0.2226
Cube-retention rate	93.86%
Failure rate	6.14%
Mean episode length	18.82 s
Position error	0.01148
Tilt error	0.2128
Fingertip contact fraction	0.2800
Fingertip contact count	1.1198
Object linear speed	0.02867
Torque L2	0.06432
Work L2	1.0526

3. Modification-1

For my first modification, I selected Track A: Reward Design and added an explicit fingertip contact reward. The baseline measured fingertip contact as an evaluation metric but did not directly reward it. I added the fingertip_cube_contact_fraction reward with weight 0.1:

$$r_{contact} = 0.1 \frac{\min(n_{contact}, 3)}{3}.$$

The reward increases with the number of fingertips touching the cube and saturates when three or more fingertips are in contact. My hypothesis was that encouraging multi-fingertip contact would produce a more stable grasp, improve cube retention and reduce pose deviation failures. A possible tradeoff was that excessive emphasis on maintaining contact could restrict the finger repositioning required for rotation.

All other settings were held equal to the baseline: 4096 environments, seed 42, the same PPO and network configuration and 5000 iteration training budget.

4. Modification-1 Result

The contact reward modification successfully changed the behavior targeted by the new reward. Compared with the baseline, mean fingertip-contact fraction increased by 33.2% from 0.2800 to 0.3729 and mean contact count increased from 1.12 to 1.49. However, this did not improve overall manipulation performance. Rotation progress decreased by 30.3% while retention declined from 93.86% to 92.22%. Position error increased by 9.1% tilt error increased by 17.7% and torque usage increased by 25.3%.

Therefore, the original hypothesis was only partially supported. The reward successfully encouraged more fingertip contact but greater contact alone did not guarantee a stable or useful grasp for rotation. The policy may have prioritized maintaining contact over repositioning its fingers effectively, producing a tradeoff between contact and rotation quality. The run is available at:
https://wandb.ai/hsharma9-university-of-southern-california/mjlab/runs/fq9ecepk?nw=nwuserhsharma9

Metric	Baseline	Contact Reward	Change from baseline
Rotation progress	0.2226	0.1551	30.3% worse

Cube retention rate	93.86%	92.22%	1.63 percentage points worse
Failure rate	6.14%	7.78%	1.63 percentage points worse
Mean episode length	18.82 s	18.56 s	1.4% worse
Position error	0.01148	0.01252	9.1% worse
Tilt error	0.2128	0.2505	17.7% worse
Contact fraction	0.2800	0.3729	33.2% better
Contact count	1.1198	1.4916	33.2% better
Linear speed	0.02867	0.02948	2.8% worse
Torque L2	0.06432	0.08058	25.3% worse
Work L2	1.0526	1.0899	3.5% worse

5. Modification-2

For my second modification, I selected Track B: Observation Design and reduced the actor's observation history length from 10 steps to 5 steps. This reduced the amount of temporal context supplied to the policy by 50% while leaving the critic, reward function, PPO configuration, network architecture, training seed and 5000 iteration budget unchanged.

My hypothesis was that a shorter observation history could remove redundant temporal information, reduce the actor input size and make policy learning more efficient. However, it could also make it harder for the policy to infer motion under observation noise and one-step sensor delay randomization.

6. Modifications-2 Results

History-5 achieved a rotation progress of 0.1853 which was 16.8% lower than the baseline but higher than the Contact Reward result. Its contact fraction improved by 8.9% from 0.2800 to 0.3049. However, retention decreased from 93.86% to 92.91%, position error increased by 7.0% and tilt error increased by 9.7%. The work metric increased by 27.9% indicating less efficient actuation.

These results did not support the hypothesis that five history steps would provide sufficient temporal context more efficiently. The longer baseline history likely helped the policy infer cube and finger motion under noisy or delayed observations. Nevertheless, History-5 provided a better overall balance than Contact Reward because it retained more rotation progress and produced lower pose error.

The run is available at
https://wandb.ai/hsharma9-university-of-southern-california/mjlab/runs/hvulk2gt?nw=nwuserhsharma9

Metric	Baseline	History-5	Change from baseline
Rotation progress	0.2226	0.1853	16.8% worse
Cube retention rate	93.86%	92.91%	0.95 percentage points worse
Failure rate	6.14%	7.09%	0.95 percentage points worse
Mean episode length	18.82 s	18.66 s	0.8% worse
Position error	0.01148	0.01228	7.0% worse

Tilt error	0.2128	0.2335	9.7% worse
Contact fraction	0.2800	0.3049	8.9% better
Contact count	1.1198	1.2194	8.9% better
Linear speed	0.02867	0.03148	9.8% worse
Torque L2	0.06432	0.07284	13.2% worse
Work L2	1.0526	1.3458	27.9% worse

7.Overall Discussion

The baseline was the best overall approach. It achieved the highest rotation progress and retention while producing the lowest pose errors, failure rate, torque and work. Contact Reward produced the highest fingertip contact but demonstrated that optimizing an intermediate behavior does not necessarily improve the full task. History-5 was the stronger of the two modifications overall, but its reduced temporal context still degraded rotation, stability and efficiency relative to the baseline.

A primary limitation is that each policy was trained using only one seed and evaluated using one fixed evaluation seed. The experiments were also conducted entirely in simulation. With additional compute, I would repeat training across multiple seeds, evaluate confidence intervals, and test alternative contact rewards that are conditioned on stable rotation rather than contact alone. I would also evaluate intermediate history lengths such as seven or eight steps to identify whether some input reduction is possible without losing important temporal context.

8. Policy Videos

1. Baseline
https://drive.google.com/file/d/1xNDfUO_Kg5yDPauNrz_iNo5H9VQ1OWlB/view?usp=sharing

2. Contact Reward
https://drive.google.com/file/d/1OJwWa7z6S4jx_ew3RVomGRslNnl-E1Ok/view?usp=sharing

3. History-5
https://drive.google.com/file/d/1qMrUZvjPJYnJ_4iVI9dgF4S2PbDam6KV/view?usp=sharing

9. Comparison Table

Metric	Baseline	Contact Reward	History-5
Rotation progress	0.2226	0.1551	0.1853
Cube retention rate	93.86%	92.22%	92.91%
Failure rate	6.14%	7.78%	7.09%
Mean episode length	18.82	18.56	18.66
Fingertip contact count	1.1198	1.4916	1.2194
Fingertip contact fraction	0.2800	0.3729	0.3049
Fingertip found maximum	0.7877	0.8516	0.7976
Linear speed	0.02867	0.02948	0.03148
Position error	0.01148	0.01252	0.01228

Tilt error	0.21277	0.25047	0.23348
Torque L2	0.06432	0.08058	0.07284
Work L2	1.05262	1.08987	1.34583